

INTRODUCTION

CHAPTER I

INTRODUCTION

1.0 Introduction

Education is a fundamental human right and a vital means of promoting individual development, social participation, and national progress. For children with disabilities, education can be truly effective only when instructional practices are tailored to their unique needs, strengths, and learning capacities. Students with multiple disabilities often encounter more complex challenges than those with a single disability, as they may experience overlapping difficulties related to communication, mobility, sensory processing, cognitive functioning, and social engagement. Consequently, they require teaching approaches that are adaptable, meaningful, interactive, and easily accessible to support their learning and overall development.

Science education is an important part of the school curriculum because it develops observation, reasoning, problem-solving, and decision-making skills. It also helps learners understand their environment, health, technology, and daily life. However, science is often taught in abstract ways, which may not be suitable for students with multiple disabilities. Many such students find it difficult to understand scientific concepts through lecture-based or text-heavy teaching. In this context, storytelling can serve as an effective teaching strategy because it presents information in a simple, meaningful, and emotionally connected manner.

Storytelling is not only an entertainment tool but also an educational method that can promote attention, comprehension, memory, and participation. In inclusive classrooms, storytelling can

help students connect new knowledge with familiar experiences. When it is combined with pictures, actions, objects, gestures, sound, and repetition, it becomes even more suitable for learners with diverse needs. The present study is concerned with the use of storytelling as a tool to enhance science knowledge among students with multiple disabilities at the secondary level.

1.1 Multiple Disabilities

Multiple disabilities refer to the presence of two or more disabilities in the same individual. These disabilities may occur in different combinations, such as intellectual disability with visual impairment, hearing impairment with cerebral palsy, or autism with communication and cognitive difficulties. The combined effect of these disabilities is usually greater than the effect of one disability alone.

Children with multiple disabilities generally experience complex educational needs. They may have difficulty speaking, understanding, moving, seeing, hearing, concentrating, or interacting with others. Because of this, they often need individualized instruction, assistive technology, structured routines, and repeated learning experiences. Their education should not be limited to academic achievement alone; it must also support communication, independence, and functional life skills.

According to special education literature, students with multiple disabilities benefit from teaching that is experiential, multisensory, and context-based. Storytelling is useful for such learners because it can simplify difficult concepts, create interest, and support learning through different modes of expression. It allows the teacher to present one concept in several forms, which makes learning more accessible.

1.2 Census Report on Disabilities

Global Context

- Global Disability Prevalence - Around 16% of the world's population ($\approx$ 1.3 billion people) live with some form of disability. This includes mild to severe functional limitations. (World Health Organization)

- Multiple Disabilities Globally: Reliable global estimates of multiple disability prevalence specifically (two or more disabilities simultaneously) vary widely by measurement method. Some country-level analyses suggest that a significant proportion of the total disabled population experience multiple disabilities.

India — Census & Survey Data National Census (India, 2011)

Total persons with disabilities in India contributes to 26.8 million (2.21% of population). Of these, 2.116 million ($\approx$ 21.16 lakh) were reported as having multiple disabilities. This means multiple disabilities made up roughly 7–8% of all persons with disability in Census data — a significant subgroup needing comprehensive support.

The 2011 census is the most recent detailed national disability survey data publicly available, but many experts believe this underestimates actual disability prevalence due to reporting gaps, stigma, and limited field screening.

1.3 Prevalence Of Children With Multiple Disabilities

The prevalence of children with special needs refers to the number or proportion of such children in the population. In many countries, including India, the actual number of children with disabilities is believed to be higher than official records show because many children remain unidentified, especially in rural and underserved areas. Reasons include lack of awareness, limited screening, stigma, poverty, and inadequate assessment services.

Children with special needs are present in both special schools and inclusive schools. Their participation in education has increased due to legal provisions, awareness programmes, and policies promoting inclusion. However, access alone does not ensure quality learning. Many children continue to face barriers in curriculum content, teaching methods, evaluation, and classroom participation.

The prevalence of special needs in school populations shows why teachers must be prepared to use effective and adaptable methods. In science education, many of these children are underserved because the subject is often considered too difficult. This makes the need for creative methods such as storytelling even more significant.

1.4 Teaching Method, Approach and Strategies

A teaching approach refers to the broad philosophical and theoretical foundation that guides instructional practices. It reflects the teacher's beliefs about the nature of learning, the role of the learner, and the purpose of education. Approaches provide the overall direction for teaching and influence the choice of methods and strategies used in the classroom. For example, learner-centered approaches emphasize active participation, experiential learning, and the construction of

knowledge through meaningful experiences, whereas teacher-centered approaches focus on direct instruction and knowledge transmission.

A teaching method is a systematic procedure or organized way of presenting subject matter to learners. Methods translate educational philosophies into practical classroom activities and provide a structured framework for instruction. Lecture, demonstration, project work, inquiry-based learning, cooperative learning, and storytelling are examples of teaching methods commonly employed in educational settings. The selection of an appropriate teaching method depends on several factors, including learning objectives, content characteristics, learner needs, classroom environment, and available resources.

Teaching strategies are specific techniques, activities, or instructional practices used within a method to enhance learning outcomes. Strategies are more flexible and situational than methods and are designed to promote learner engagement, comprehension, retention, and application of knowledge. Examples of teaching strategies include questioning, discussion, role-play, visual aids, peer tutoring, concept mapping, reinforcement, and storytelling. Effective teaching strategies help bridge the gap between instruction and learning by making educational experiences meaningful, interactive, and accessible.

In recent decades, educational research has increasingly emphasized the importance of innovative and inclusive teaching strategies that accommodate learner diversity. Traditional methods of instruction often present challenges for students with disabilities, particularly those with multiple disabilities, who may experience difficulties in communication, cognition, attention, memory, and social interaction. Consequently, educators are encouraged to adopt

learner-centered and multisensory instructional approaches that actively engage students and support individual learning needs.

Among the various instructional strategies available, storytelling has emerged as a powerful pedagogical tool. Storytelling involves the use of narratives to communicate information, experiences, values, and concepts in a meaningful and engaging manner. As an instructional strategy, storytelling creates contextualized learning experiences that connect new information with learners' prior knowledge and real-life experiences. Stories capture attention, stimulate imagination, enhance memory, and facilitate understanding of complex concepts by presenting them in a coherent and relatable format.

The educational value of storytelling is supported by theories of constructivism, social learning, and cognitive psychology, which suggest that learners construct knowledge more effectively when information is presented within meaningful contexts. Through stories, abstract concepts can be transformed into concrete experiences, making learning more accessible and enjoyable. Storytelling also promotes language development, critical thinking, emotional engagement, creativity, and social interaction, all of which contribute to effective learning.

In the field of science education, storytelling provides opportunities to present scientific concepts through narratives that are engaging, memorable, and relevant to students' everyday lives. By integrating scientific facts and principles within stories, educators can foster curiosity, enhance conceptual understanding, and improve knowledge retention. For students with multiple disabilities, storytelling can serve as an inclusive instructional strategy by combining visual,

auditory, linguistic, and experiential elements that accommodate diverse learning styles and abilities.

Therefore, the thoughtful integration of appropriate teaching approaches, methods, and strategies is essential for achieving educational goals and ensuring meaningful participation of all learners. Storytelling, as an innovative and learner-centered instructional strategy, holds considerable potential for enhancing science knowledge among students with multiple disabilities by making learning more engaging, accessible, and effective.

1.5 Teaching of Science

Science is a body of knowledge and a process of inquiry that helps us understand the natural and physical world. It is also a way of thinking, investigating, and solving problems. At the school level, science includes topics related to living organisms, matter, energy, environment, health, and technology. It helps learners develop curiosity, critical thinking, and practical understanding.

The teaching of science should not be limited to memorizing facts. Instead, it should encourage students to observe, ask questions, compare, classify, experiment, and draw conclusions. In secondary education, science becomes more detailed and concept-oriented, which makes it challenging for many learners, particularly those with disabilities. Concepts such as digestion, force, evaporation, reproduction, and electricity may be difficult to understand when taught only through textbooks.

For students with multiple disabilities, science teaching must be made meaningful and concrete. Teachers should use examples from daily life, demonstrations, models, pictures, real objects, and simple explanations. Storytelling can help by turning abstract information into a sequence of understandable events. For example, instead of only explaining the water cycle, a teacher can tell a story about a drop of water traveling from a pond to the sky and back to the earth. Such methods make science more engaging and memorable.

1.6 Teaching Science at secondary level

Teaching science to children with multiple disabilities requires careful planning and adaptation. Many learners with disabilities face difficulties in reading, language comprehension, visual processing, attention, memory, and communication. These difficulties can make science seem inaccessible if it is taught in a traditional, lecture-centered way.

A good science teacher must therefore use multiple modes of teaching. These include demonstration, activity-based learning, simplified explanations, repetition, cooperative learning, and the use of concrete materials. For some learners, tactile experiences and visual supports are essential. For others, communication aids or alternative response modes may be necessary. The teacher must ensure that every student can participate in some form.

Students with multiple disabilities may also need more time and more support to understand science concepts. Because of this, the focus should be on functional understanding rather than only on factual recall. Storytelling is useful in this context because it reduces anxiety, increases interest, and presents science in a meaningful context. It can be used to teach everyday scientific ideas such as cleanliness, food, plants, weather, and the human body.

1.7 Storytelling

Storytelling is the act of conveying ideas, events, experiences, and knowledge through narrative form. It is one of the oldest and most natural methods of communication. In education, storytelling helps learners listen, imagine, organize information, and remember content more effectively.

Storytelling can be oral, written, visual, digital, or multisensory. In special education, multisensory storytelling is especially useful because it combines speech with pictures, gestures, real objects, sound, and movement. This supports children with limited language or cognitive abilities and makes learning more inclusive. Storytelling also creates emotional engagement, which improves attention and memory.

In science education, storytelling can transform a difficult concept into a meaningful event. For example, a lesson on plants can be presented as a story about a seed's journey to becoming a tree. Such a narrative helps students understand sequence, process, and purpose. For students with multiple disabilities, this method can improve comprehension and participation.

1.8 Storytelling in Enhancing Science Knowledge

Storytelling can enhance science knowledge by making scientific concepts more concrete, interesting, and memorable. According to educational studies, students learn better when new information is connected to a familiar context. Storytelling does this by placing science ideas within a sequence of events, characters, and actions.

This method is particularly useful for learners who struggle with reading or abstract thinking. Instead of teaching scientific facts in isolation, storytelling presents them in a connected manner. It can explain the human body, seasons, hygiene, pollution, food chains, or the water cycle through simple narratives. This helps students understand not only what happens, but why it happens.

For children with multiple disabilities, storytelling can also improve language exposure, attention span, and participation. The use of repetition, visual cues, and concrete materials makes the story accessible. In this way, storytelling becomes both a teaching method and a support for inclusive science learning.

1.9 Need and Significance of the Study

This study is needed because many students with multiple disabilities at the secondary level face serious difficulties in learning science through conventional methods. Science is often abstract, vocabulary-heavy, and concept-driven, which makes it less accessible to learners with communication, cognitive, and sensory challenges. Without adapted teaching strategies, these students may be excluded from meaningful science learning.

Storytelling offers a promising solution because it is simple, flexible, and student-friendly. It can reduce fear of difficult content, increase engagement, and improve understanding through narrative structure. It also allows the teacher to use pictures, actions, props, and repetition, which are important for special education. Therefore, the study has practical significance for improving classroom instruction.

The study is also significant because it may contribute to inclusive education practice. It can help teachers understand how storytelling can be used as an instructional tool in science. It may also support the development of teaching materials, lesson plans, and intervention strategies for students with multiple disabilities. In the long run, it may improve access to quality science education and support the academic inclusion of learners with special needs.

1.10 Statement of the Problem

Science is a dynamic and systematic discipline that helps human beings organize knowledge about the natural world through observation, questioning, experimentation, reasoning, and continuous inquiry. It develops in learners the capacity to think critically, solve problems, make informed decisions, and understand the relationship between scientific knowledge and everyday life. In the context of school education, science is not merely a subject concerned with facts and formulas; rather, it is a means of developing curiosity, creativity, innovation, and scientific temperament among students. These competencies are essential not only for academic achievement but also for future personal, social, and vocational growth.

Science education is therefore important for all learners, including students with multiple disabilities. Inclusive science learning provides opportunities for students to explore their surroundings, engage with real-life phenomena, and build essential cognitive and functional skills. However, students with multiple disabilities often encounter substantial barriers in accessing, comprehending, and retaining scientific concepts. Their learning needs may include limitations in communication, mobility, sensory processing, attention, and cognitive functioning, which can make conventional science instruction difficult and less meaningful. As a result, these

students may experience frustration, reduced participation, and limited academic progress in science.

The problem becomes more serious at the secondary level, where science content becomes more abstract, conceptually demanding, and language-intensive. At this stage, students are expected to understand complex ideas, connect concepts across topics, and apply knowledge in new situations. For students with multiple disabilities, such expectations may create a wide achievement gap in science learning when teaching methods are not sufficiently adapted. This gap often results in limited exposure to scientific knowledge, reduced confidence in learning science, and fewer opportunities for higher education and career pathways in science, technology, and related fields.

In many classrooms, science is still taught through traditional methods that rely heavily on verbal explanation, textbook reading, and rote memorization. Such approaches may not adequately address the diverse learning needs of students with multiple disabilities. When instruction does not provide enough sensory support, meaningful context, repetition, or student engagement, learners may become passive, uninterested, and disconnected from the lesson. Over time, this can lead to low motivation, poor self-esteem, and disengagement from science learning. Therefore, there is a clear need for instructional practices that are inclusive, creative, and responsive to the needs of these learners.

Storytelling has emerged as a promising pedagogical strategy in inclusive education because it can present scientific ideas in a simple, engaging, and memorable form. Through storytelling, abstract concepts can be transformed into meaningful experiences that connect with the learner's

daily life, imagination, and emotions. It can also support comprehension by organizing information into a sequence of events and by using familiar language, repetition, and multisensory supports. For students with multiple disabilities, storytelling may help reduce anxiety, increase participation, and improve understanding by making science more accessible and enjoyable.

Despite the potential of storytelling as an instructional tool, there is limited research on its effectiveness in enhancing science knowledge among students with multiple disabilities at the secondary level. Most science teaching practices continue to be based on conventional approaches that may not sufficiently address inclusive learning needs. This creates a gap between the need for accessible science instruction and the teaching methods currently used in practice. Hence, the present study seeks to investigate the effectiveness of storytelling in enhancing science knowledge among secondary students with multiple disabilities and to contribute to the development of inclusive science education practices.

1.11 Definitions

- **Storytelling**

Storytelling is the process of using narrative to communicate experiences, values, beliefs, and knowledge in a meaningful way. (Lambert, J. (2019)

Storytelling is a mode of thought and a vehicle for meaning-making through which people organize and interpret their experiences. (Bruner, J. (1991)

Storytelling is the act of using stories to communicate information, knowledge, wisdom, and values. (Haven, K. (2022))

- **Science**

Science is the complete and consistent description of the facts of experience in the simplest possible terms. (Pearson)

Science is the systematic enterprise that builds and organizes knowledge in the form of testable explanations. (UNESCO Science Report)

Science is the use of evidence to construct testable explanations and predictions of the natural phenomenon, as well as the knowledge generated through the process. (National Academy of Sciences, 1998)

- **Multiple Disabilities**

Multiple disabilities means concomitant impairments (such as intellectual disability-blindness, intellectual disability-orthopedic impairment, etc.), the combination of which causes such severe educational needs that they cannot be accommodated in special education programs solely for one of the impairments. (IDEA)

Multiple disabilities means more than one of the specified disabilities, including deafblindness, where a person experiences a combination of disabilities causing substantial limitations. (RPwD Act, 2016)

- **Secondary Level**

Secondary education is designed to complete the provision of basic education and to lay the foundation for lifelong learning and human development. (ISCED)

Secondary education refers to the stage of schooling that follows elementary education, preparing learners for higher secondary education and vocational pathways. (NCERT).

Secondary education is the stage of education between elementary and higher education that aims at the intellectual, physical, social, and emotional development of adolescents. Aggarwal, J.C. (2009).

1.12 Objectives of the Study

- To analyse the science knowledge among students with multiple disabilities at secondary level.
- To develop and implement storytelling sessions to enhance science knowledge among students with multiple disabilities at secondary level.

- To check the effectiveness of storytelling in enhancing science knowledge among students with multiple disabilities at secondary level.

1.13 Hypotheses

- The use of storytelling significantly improves the knowledge of science among students with multiple disabilities at secondary level.
- The use of storytelling significantly improves the knowledge of science with respect to cognitive, affective, psychomotor and application domains among students with multiple disabilities at secondary level.

1.14 Scope of the Study

This study focuses on the use of storytelling as a teaching strategy to enhance science knowledge among students with multiple disabilities at the secondary level. It examines whether storytelling, supported by visual flash cards and real food items, can improve students' understanding of a selected science concept in a more engaging and meaningful way.

The study is limited to secondary-level students with multiple disabilities who need concrete, multisensory, and individualized instruction. It considers their learning difficulties, participation, attention, and response during the storytelling-based science intervention. The study also explores how the selected instructional method supports comprehension and retention of the science concept.

The content of the study is restricted to one selected topic related to food. The intervention is designed to teach this concept through the story “The Magic Kitchen.” Therefore, the study does not cover the entire science curriculum, but only the chosen concept.

The study includes the use of storytelling as the main teaching method and supporting materials such as flash cards, pictures, and real food samples. It focuses on the learning outcome in terms of science knowledge gained after the intervention. The findings may be useful for special educators, science teachers, and researchers working in inclusive education.

1.15 Delimitations

The study is delimited to students with multiple disabilities at the secondary level. It does not include students at primary or higher secondary level, or students without disabilities. This makes the study specific to a particular learner group and age range.

The study is limited to one instructional strategy, namely storytelling. It does not compare storytelling with other teaching methods such as lecture, demonstration, project method, or activity-based learning. The focus is only on the effectiveness of storytelling in improving science knowledge.

The study is also delimited to one science topic. It does not assess science learning in other areas such as plants, water, matter, energy, or the human body. This narrow focus helps in studying the intervention clearly and in depth.

The intervention is carried out using visual flash cards and real food items only. Other teaching aids, technologies, or digital tools are not included in the study. The research is further limited to the specific time period and setting in which the intervention was conducted.

Thus, the study is designed with clear boundaries in terms of participants, content, method, and materials. These delimitations help maintain the focus of the research and make the results more specific to the selected context.

REVIEW OF RELATED LITERATURE

CHAPTER- II

REVIEW OF RELATED LITERATURE

2.1 Introduction:

This chapter includes a review of the literature as well as an overview of the research project. It has been divided into different sections based on the study's aims. The results from the past to the present have been presented in the literature review. The review of previous literature is organized around the study's variables. The methodical identification, location, and analysis of materials providing information connected to the study issue constitute the review of related literature.

The fundamental goal of reviewing the literature is to figure out what has already been done on the subject. Previous research provides support for the current research idea, as well as suggestions for what needs to be done and the importance of the study. The general review summarizes what has been accomplished and what remains to be accomplished. Another important goal of reviewing the literature is to identify research strategies and specific data collection approaches that have or have not been successful in previous studies on topics similar to the one under consideration. This information aids the researcher in making the most of their experiences. Also, suggest new ideas and procedures that haven't been examined before. A literature review provides the reader with up-to-date information on the status of research in this discipline and introduces the reader to any opposing ideas and viewpoints on the subject.

There are four basic goals of a literature review:

1. It examines the literature in your field of study. It compiles a summary of the information found in the literature.
2. It critically evaluates the data acquired by identifying knowledge gaps, demonstrating the limitations of ideas and points of view, identifying opportunities for further research, and analyzing areas of contention. It presents the literature in a systematic manner.
3. A literature review demonstrates to readers that you have a thorough understanding of your subject and that you recognize how your study fits into and contributes to an existing body of knowledge. Here's another way to put those four major responsibilities. Review of the literature.
4. Improves the legitimacy of work by demonstrating familiarity with a body of knowledge; summarizes previous research and explains how it relates to your topic; demonstrates that you have learned from others and that your research is a starting point for new ideas; integrates and summarizes.

2.2 Importance of literature review:

The importance of literature review in scientific manuscripts can be condensed into an analytical feature to enable the multifold reach of its significance. It adds value to the legitimacy of the research in many ways:

1. Provides the interpretation of existing literature in light of updated developments in the field to help in establishing the consistency in knowledge and relevancy of existing materials.

2. It aids in calculating the impact of new information in the field by charting their knowledge progression.

3. It establishes facts by bringing out the dialectics of inconsistencies between distinct concepts within the field.

4. The research gaps that were initially examined are subsequently investigated in order to establish the most recent facts and theories in order to offer value to the field.

5. Checks the research for relevance and coherency by providing information.

2.3 The review of literature studies is given under the following sub-headings: -

- Research studies related on storytelling for students with disabilities
- Research Studies related to enhance science knowledge among students with Multiple Disabilities
- Research Studies related to storytelling to enhance science knowledge among students with Multiple Disabilities

2.3.1 Research studies on storytelling.

- **Mims et al. (2019)** conducted a study entitled *"Increasing Comprehension of Students with Significant Intellectual Disabilities through Story-Based Lessons."* The study investigated whether story-based lessons could improve reading comprehension among students with significant intellectual disabilities. The sample consisted of

elementary and middle school students with intellectual disabilities. Story-based lessons incorporated adapted texts, repeated readings, and interactive questioning. Results indicated substantial gains in comprehension and participation. The researchers concluded that storytelling can facilitate learning and communication among students with significant disabilities and recommended its use across academic subjects.

- **Delano and Snell (2020)** conducted a study entitled *"The Effects of Social Stories on the Social Engagement of Children with Autism."* The study examined the effectiveness of narrative-based interventions in improving social behaviors among children with autism. Participants included elementary-aged students diagnosed with ASD. Social Stories were used to teach appropriate social skills and behaviors. Results demonstrated improvements in social engagement, communication, and peer interactions. The researchers recommended using storytelling as a practical intervention for enhancing social understanding among students with autism.
- **Isbell et al. (2024)** conducted a study titled *"The Effects of Storytelling and Story Reading on the Oral Language Complexity and Story Comprehension of Young Children."* The study explored how storytelling influences language development. The sample included young children, including those with developmental delays. Students participated in storytelling and story-reading activities over several weeks. Findings showed significant improvements in vocabulary development, oral language complexity, and story comprehension. The authors suggested that storytelling should be integrated into educational programs for children with developmental disabilities.

- **Miller and Pennycuff (2023)** conducted a study entitled *"The Power of Story: Using Storytelling to Improve Literacy Learning."* The study investigated storytelling as a strategy for improving literacy among diverse learners, including students with learning difficulties. Participants included elementary school students receiving literacy support services. Results showed increased motivation, listening skills, vocabulary acquisition, and reading comprehension. The researchers recommended storytelling as an inclusive instructional strategy that can support students with varying learning needs.
- **Browder et al. (2018)** conducted a study entitled *"Teaching Elementary Students with Multiple Disabilities to Participate in Shared Stories."* The purpose of the study was to investigate whether adapted shared-story interventions could improve participation and communication among students with multiple disabilities. The sample consisted of elementary-aged students with severe and multiple disabilities. Teachers used adapted books, tactile supports, repeated storytelling, and interactive communication opportunities. Results indicated increased student participation, attention, communication responses, and engagement during story sessions. The researchers recommended incorporating multisensory storytelling techniques into special education curricula to promote learning and communication.
- **Scruggs and Mastropieri (2016)** conducted a study titled *"Classroom Applications of Mnemonic Instruction: Acquisition, Maintenance, and Generalization."* The study examined the effectiveness of mnemonic instruction in teaching science content to students with mild disabilities. Nineteen students with mild disabilities participated in a within-subjects experimental design. Students received both mnemonic and

traditional instruction in life science units. Results indicated that mnemonic instruction significantly improved acquisition, retention, and generalization of science information compared to traditional teaching methods. The researcher recommended integrating mnemonic strategies into science instruction for students with disabilities.

- **Phillips, L (2018)** Conducted a study on *Storytelling-The Seeds of Children's Creativity*. This study describes a storytelling-based educational curriculum. According to the study, storytelling can help young children create a stronger sense of community, improve knowledge and memory recall, assist early literacy development, and increase their creative potential. Through a variety of extension initiatives, the study investigated storytelling's potential for this. Finally, it is argued that storytelling plays an extremely important function in the teaching of young children.

2.3.2 Reviews based on enhancing science knowledge among Children with Multiple Disabilities

- **Apanasionok et al. (2019)** conducted a systematic review titled *"Teaching Science Skills and Knowledge to Students with Developmental Disabilities."* The review examined 30 intervention studies and 20 studies exploring teacher and student experiences. Participants included students with intellectual disabilities and autism spectrum disorders, many of whom had additional disabilities. Results indicated that systematic instruction was the most effective method for teaching science vocabulary, concepts, inquiry skills, and comprehension skills. Students and teachers reported

positive learning experiences. The authors recommended investigating science instruction for students with more severe and multiple disabilities and focusing on complex science concepts.

- **Brown (2015)** examined the role of tactile-visual integration in a study titled *Role of tactile-visual integration in teaching basic science concepts to children with multiple disabilities*. The sample consisted of 20 students with multiple disabilities. Instruction involved the use of textured materials, three-dimensional models, and visual representations. The findings indicated significant improvement in concept comprehension and sensory coordination. The study concluded that tactile-visual integration is a crucial component of effective multisensory science instruction
- **Iatraki and Soulis (2021)** conducted a review entitled *"A Systematic Review of Single-Case Research on Science-Teaching Interventions to Students with Intellectual Disability or Autism Spectrum Disorder."* The researchers analyzed 24 single-case studies involving students with intellectual disabilities and autism. The review found that science interventions significantly improved science vocabulary, concept acquisition, and inquiry skills. Visual supports, systematic instruction, and repeated practice were identified as highly effective teaching strategies. The study recommended extending these interventions to students with severe cognitive impairments.
- **Mikropoulos and Iatraki (2023)** conducted a systematic review titled *"Digital Technology Supports Science Education for Students with Disabilities."* The review analyzed 21 empirical studies published between 2013 and 2021. Participants included students with intellectual disabilities, autism, developmental disabilities, and multiple

disabilities. Findings revealed that digital technologies such as virtual reality, augmented reality, tablets, and multimedia applications increased student motivation, engagement, and science achievement. The researchers recommended integrating digital tools into science instruction and investigating technology-based interventions for students with disabilities.

- **Spooner et al. (2021)** conducted a review titled *"Evaluating Evidence-Based Practice in Teaching Science Content to Students with Severe Developmental Disabilities."* The study reviewed 17 experimental studies published between 1985 and 2009 that focused on science instruction for students with severe developmental disabilities, including students with multiple disabilities. Fourteen studies were identified as high- or adequate-quality research. Results showed that systematic instruction, task analysis, prompting, and inquiry-based adaptations effectively improved science knowledge and participation. The authors recommended expanding evidence-based science instruction and increasing access to science education for students with severe disabilities.
- **Alvarez Ariza and Hernandez Hernandez (2025)** conducted a systematic literature review titled *"Research-Based Interventions and Strategies for Students with Disabilities in STEM and STEAM Education."* The review synthesized research on interventions designed to improve participation and achievement in science, technology, engineering, and mathematics among students with disabilities. Findings indicated that hands-on activities, assistive technology, inquiry-based learning, and differentiated instruction positively influenced science learning outcomes. The authors recommended the development of inclusive STEM programs that specifically address the needs of students with disabilities.

- **Brigid Barron & David Osher (2020)** drawn the implications for school and classroom practices of an emerging consensus about the science of learning and development, outlined in a recent synthesis of the research. Situating the review in a developmental systems framework, we synthesize evidence from the learning sciences and several branches of educational research regarding well-vetted strategies that support the kinds of relationships and learning opportunities needed to promote children's well-being, healthy development, and transferable learning. In addition, we review research regarding practices that can help educators respond to individual variability, address adversity, and support resilience, such that schools can enable all children to find positive pathways to adulthood.
- **Mehta (2019)** conducted an experimental study titled *Effectiveness of multisensory science kits for children with multiple disabilities*. The sample consisted of 26 students with multiple disabilities. The multisensory science kits included visual charts, textured materials, sound recordings, and hands-on experiments. Analysis of the results indicated significant improvement in concept understanding and retention. The study concluded that multisensory science kits are effective instructional tools for teaching basic science concepts to children with multiple disabilities
- **Tessori (2023)** conducted an experimental study titled *Multisensory learning approach for teaching science to children with multiple disabilities*. The sample included 20 children with intellectual and sensory impairments. Instruction involved visual, tactile, auditory, and kinesthetic activities to teach basic science concepts. The pre-test and post-test analysis revealed significant improvement in concept understanding and sensory engagement. The study concluded that multisensory

learning enhances concept clarity and retention among children with multiple disabilities.

- **Dunn and Dunn (2018)** conducted a study titled *Effectiveness of instructional strategies in science education for children with multiple disabilities*. An experimental design was used with a sample of 25 students from special schools. Multisensory techniques such as visual aids, auditory input, hands-on activities, and movement-based learning were implemented. The findings indicated improved attention, participation, and academic performance. The study concluded that multisensory instruction effectively supports diverse learning needs.
- **Kumar and Sharma (2023)** conducted an experimental study titled *Effectiveness of multisensory methods in teaching basic science concepts to children with multiple disabilities*. The study aimed to assess the impact of multisensory instructional methods on science learning. The sample consisted of 30 children with multiple disabilities. Multisensory teaching kits comprising real objects, pictures, sounds, and physical activities were used during instruction. The post-test results revealed significant improvement in students' understanding and application of basic science concepts. The study concluded that multisensory methods enhance learning effectiveness among children with multiple disabilities.
- **Patel (2018)** carried out a study titled *Impact of multisensory learning activities on science concept acquisition among children with multiple disabilities*. A pre-test–post-test research design was employed with a sample of 22 students. The intervention involved tactile exploration, visual demonstrations, auditory explanations, and movement-based activities. The findings indicated increased engagement, improved

attention span, and enhanced learning outcomes. The study concluded that multisensory learning activities promote active participation and effective learning among children with multiple disabilities.

- **Rao and Singh (2019)** conducted an experimental study titled *Retention of science concepts among children with multiple disabilities*. The sample comprised 24 students with multiple disabilities. The intervention integrated visual aids, auditory instructions, tactile experiences, and kinesthetic activities. The findings revealed improved retention and recall of basic science concepts. The study concluded that multisensory teaching strategies enhance memory and learning outcomes among children with multiple disabilities.
- **Hallahan, Kauffman, and Pullen (2024)** conducted a study titled *Sensory integration strategies in science education for children with multiple disabilities*. The sample consisted of 30 students with multiple disabilities. Sensory-based activities were systematically incorporated into science lessons to support concept learning. The findings indicated enhanced understanding, improved attention, and increased classroom participation. The study concluded that sensory integration strategies play a vital role in multisensory science education for children with multiple disabilities.

2.3.4 Research Studies related to storytelling to enhance science knowledge among students with Multiple Disabilities

- **Almumen and Al-Jafar (2025)** investigated *the impact of storytelling on science concept acquisition among students with severe intellectual disabilities*. The

intervention combined storytelling, digital technology, hands-on activities, and real-life observation of plant growth. Findings revealed substantial improvement in students' understanding of botany concepts and their ability to describe scientific phenomena. The study concluded that storytelling can be an effective medium for teaching science concepts to learners with significant disabilities.

- **ten Brug et al. (2016)** conducted a comparative study on *multisensory storytelling and regular storytelling among individuals with profound intellectual and multiple disabilities*. The researchers found that multisensory storytelling significantly increased learners' attention and participation compared to traditional storytelling methods. The use of individualized sensory stimuli helped learners remain engaged for longer periods and improved their responsiveness to educational content. The study highlighted the effectiveness of storytelling as an inclusive teaching strategy for learners with multiple disabilities and storytelling can help students with multiple disabilities develop sustained attention, which is essential for learning science and other academic subjects
- **Penne et al. (2022)** examined *the role of staff interaction during storytelling sessions for individuals with profound intellectual and multiple disabilities*. The study found that supportive and responsive interactions by educators enhanced learner engagement and participation during storytelling activities. The researchers emphasized that storytelling creates meaningful communication opportunities and promotes active learning among students with complex disabilities
- **Engel et al. (2018)** examined *storytelling as a strategy for science education*. The authors reported that storytelling increased students' curiosity, engagement, and

understanding of scientific concepts. By presenting scientific information in narrative form, students were able to connect abstract ideas with real-life situations, resulting in deeper comprehension and retention of knowledge.

- In her doctoral research, **ten Brug (2017)** examined the broader *educational value of storytelling for persons with profound intellectual and multiple disabilities*. The study found that storytelling not only increased enjoyment and participation but also supported learning, communication, environmental awareness, and social interaction. The findings suggest that storytelling can be used as an effective instructional tool across different learning domains, including science education.
- **McLaughlin et al. (2018)** examined the *impact of sustained science engagement activities including storytelling among secondary school students*. The study found improvements in students' interest, engagement, awareness, and understanding of science topics. Although the study was not conducted specifically with students with multiple disabilities, it demonstrated that engaging and interactive approaches, including narrative-based methods, can significantly enhance science learning outcomes.
- **Verma (2020)** conducted an experimental study titled *Effectiveness of multisensory instruction on science learning among children with multiple disabilities*. The sample included 28 students with multiple disabilities. Multisensory instructional strategies involving visual, auditory, and kinesthetic activities were implemented. Post-test analysis revealed significant improvement in academic achievement and learner engagement. The study concluded that multisensory instruction is highly effective in teaching science concepts to children with multiple disabilities.

2.4 Summary of Review

The reviewed studies indicate that storytelling, enhances attention, alertness, engagement, communication, concept understanding, and knowledge retention among learners with intellectual and multiple disabilities. Studies related to science education further demonstrate that storytelling helps students understand complex scientific concepts through meaningful and contextualized learning experiences. However, very limited research has specifically investigated the effectiveness of storytelling in enhancing science knowledge among secondary-level students with multiple disabilities, thereby establishing the need for the present study.

METHODOLOGY

CHAPTER III

METHODOLOGY

Methodology with effective procedure is the basis necessity in any research study. The success of any research depends upon the kind of methodologies and procedures followed in the steps wise execution of the study by the researcher.

This chapter outlines the research design, methodology, and procedures used to investigate the effectiveness of storytelling method in enhancing science knowledge among students with multiple disabilities attending secondary school. The methodological framework includes the research design, population, sampling, tool development, intervention structure, materials, and ethical procedure. By following a structured and systematic approach, the study ensures credible, reliable, and valid results that contributes meaningfully in the enhancement of science knowledge. Pre and post-test experimental design was employed to evaluate the impact of storytelling method in enhancing science knowledge.

3.1 Research Design

A research design is the plan or framework used to conduct a research study. It involves outlining the overall approach and methods that will be used to collect and analyse data to answer research questions or test hypotheses. Well-designed research study should have a clear and well-defined research question, a detailed plan for collecting data, and a method for

analysing and interpreting the results. A well-thought-out research design addresses all these features.

The study employs pre-test-post-test experimental design. This design allows for the comparison of knowledge in science in students before and after the introduction of the story-telling method, serving as the pre-test-post-test experimental intervention.

Pre-test establishes a baseline for each participant's knowledge in science concept, while the post-test measures the effectiveness of the intervention.

3.2 Variables

3.2.1 Definition of Variables

A variable is any characteristic, number, or quantity that can be measured or categorized. In research, variables play a crucial role in establishing relationships between different factors under study, Variables can change or vary, making them essential elements in experimental research.

3.2.2 Types of Variables

1. Independent Variable- The independent variable is the factor that is manipulated or controlled by the researcher to observe its effect on the dependent variable. It is also known as the predictor variable. The independent variable for this study is Story-telling method.

2. Dependent Variable- The dependent variable is the outcome or response that is measured in the study. It is expected to change due to the manipulation of the independent variable. The dependent variable for this study is science knowledge.

3.2.3 Variables in the Present Study

In this research on using a multisensory method to enhance the basic science concepts among children with multiple disabilities, the variables are as follows

Independent Variable: Storytelling method

Dependent Variable: Science knowledge

3.3 Area, Population and sampling

The study population consists of six children (6 males) for the age group 11 to 14 years, all attending the National Institute for the Empowerment of Persons with Multiple Disabilities (Divyangjan) (NIEPMD) model school Chennai, which caters to children with multiple disabilities. The selection criteria were based on the following:

Age-Participants were between 11 and 14 years old

Diagnosis- All participants have been diagnosed with multiple disabilities (ASD+ID)

Table 1 Distribution of the Subjects:

S.no	Name	Age	Gender	Type of disability
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1.	Subject 1	13	Male	ASD+ID
2.	Subject 2	12	Male	ASD+ID
3.	Subject 3	12	Male	ASD+ID
4.	Subject 4	13	Male	ASD+ID
5.	Subject 5	14	Male	ASD+ID
6.	Subject 6	14	Male	ASD+ID

3.4 Sampling Techniques

3.4.1 Definition of Sampling

Sampling refers to the process of selecting a subset of individuals from a larger population to participate in a study. The selected sample should represent the population to ensure the generalizability of findings.

3.4.2 Types of Sampling Techniques

1. Probability Sampling

In probability sampling, every member of the population has an equal chance of being selected. Common types include:

a. Simple Random Sampling- Each participant is chosen randomly, ensuring an equal selection probability.

b. Stratified Sampling- The population is divided into subgroups (strata), and samples are taken from each group.

c. Systematic Sampling-Participants are selected at regular intervals from an ordered list.

d. Cluster Sampling-The population is divided into clusters, and entire clusters are randomly selected.

2. Non-Probability Sampling

In non-probability sampling, participants are selected based on non-random criteria, making it less generalized. Common types include:

a. Convenience Sampling- Participants are chosen based on availability and willingness to participate.

b. Purposive Sampling-Participants are selected based on specific characteristic relevant to the study.

c. Snowball sampling- Existing participants help recruit more participants.

d. Quota Sampling- Participants are selected based on predefined characteristics in a set proportion.

3.5 Sampling in the present Study

This study follows a **Purposive Sampling technique**, a type of non-probability sampling. The sample is selected based on specific criteria that aligns with the research objectives.

3.6 Inclusion Criteria

1. Children aged 11-14 years.
2. Enrolled in the NIEPMD Model School for children with multiple disabilities.

3.7 Exclusion Criteria

1. Children below 11 and above 14 years of age.

3.8 Duration of the study

The study was conducted over a period of six months. The duration was selected to ensure sufficient time for introduction, practice, and assessment of the story-telling method, allowing for enhancing science knowledge.

3.9 Research Tool

Research Tools can be defined as vehicles that broadly facilitate research and related activities. Research Tools enable researchers to collect, organize, analyse, visualize, and publicized research outputs.

Research Tools consists of consists of hierarchical set of nodes. It has four main nodes:

1. Searching the literature,

2. Writing a paper,
3. Targeting suitable journals, and
4. Enhancing visibility and impact

For the present study the investigator used the following tools

1. Personal Proforma Sheet (Appendix 1)

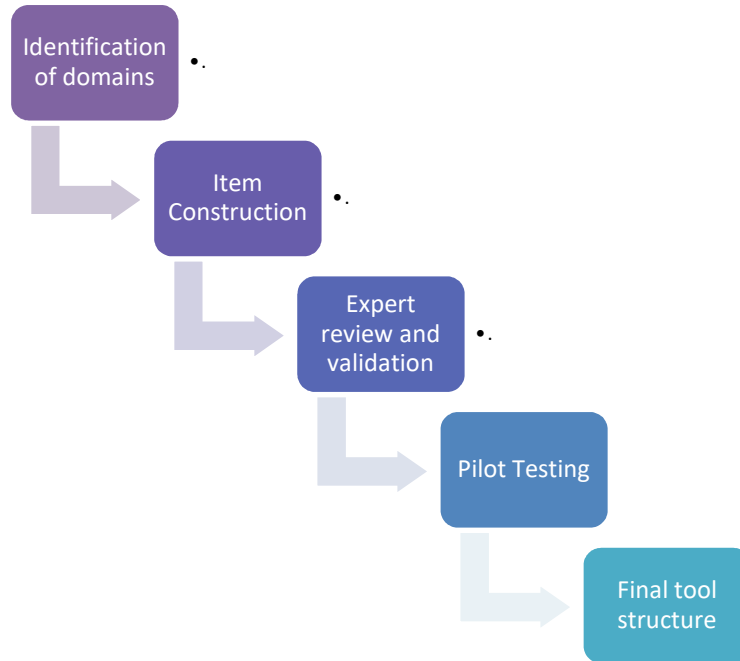
Appendix 1 consists of a Personal Proforma Sheet of the child to collect the child's demographic information. Demographic information includes the child's name, gender, date of birth, age, no. of siblings, mother's name, father's name, community, disability diagnosis, severity level of the disability, mother tongue, mode of communication. It gives all the information about the subject that is required for the research.

2. Food Awareness and Recognition Assessment Tool (Appendix 2)

The food awareness and recognition assessment tool used in the present study was developed based on established frameworks including Bloom's Taxonomy. The development process involved several phases: domain identification, item construction, expert validation and revision

3.10 Phases of Development of the tool

Figure 1 Phases of Development of tool



Phase 1: Identification of Domains:

The first phase involved an extensive review of literature, existing assessment tools, frameworks relevant to science knowledge among students with multiple disabilities. Based on the review and specific needs of the target population, the following four domains were selected for inclusion in the tool:

I. Cognitive

II. Affective

III. Psychomotor

IV Application

Phase 2: Item Construction:

Based on the identified domain, a pool of items was drafted drawing from established tools such as Bloom's Taxonomy. The items were carefully constructed to match the cognitive levels of secondary-aged children with multiple disabilities.

Initially, the tool comprised a tool 20 items with 3-point rating scale

I. Cognitive : 6

II. Affective : 4

III. Psychomotor : 6

IV Application : 4

Figure 3.2-Description of the Rating Scale

Achieved -The student consistently demonstrates science knowledge through storytelling method
Emerging - The student sometimes demonstrates science knowledge through storytelling method on some occasions, but not consistently.
NA - The student never demonstrates science knowledge through story-telling method

Phase 3: Expert review and validation

The validity and reliability of a research tool are critical to ensuring the accuracy and consistency of the results. The tool used in this study was validated through 10 expert reviews and pilot testing.

In this phase, the draft version of the tool was shared with panel of professionals including special educators, rehabilitation officer, psychologists working in the field of multiple disabilities. These experts were asked to review the tool for content relevance, clarity of items, developmental appropriateness, and alignment with practical use in real settings.

Based on their detailed feedback, revisions were made to improve the wording of items. move redundancies, and ensure linguistic suitability. Items that were unclear or overlapped with others were either rephrased or eliminated.

After expert review:

Final 24 items were drafted in the areas of Cognitive, Affective, Psychomotor and Application.

9 items modified for clarity and relevance.

4 items were removed due to low relevance (due to overlap).

Finally, 20 items were drafted

This phase helped strengthen the content validity of the tool by ensuring that each item meaningfully contributed to assessing the intended cognitive, affective, psychomotor and application domain.

Phase 4: Pilot testing:

The purpose of the pilot study was to see if the tool is understandable, relevant, consistent and self-defined enough to meet the study hypothesis. A pilot-testing was conducted on 2 samples Multiple Disabilities attending secondary level. Prior permission was taken from the Head of the Model School under NIEPMD. Informed consent was taken from the teacher and the parent in the pilot study. The pilot study helped the researcher in finishing the structure, format, administrator, and intervention procedure of the research.

3.11 Reliability

Reliability in general refers to the consistency and dependability of measure or process. Reliability defined as the extent to which results are consistent over time (Joppe, 2000).

3.11.1 Types of Reliability

Test-retest reliability- This refers to the consistency of results when a test is administered to same individuals at different times.

Parallel forms reliability- This strategy involves giving the same group of people multiple types of tests to determine if the results stay the same way when using different research methods

lateral consistency- this assesses how well the items within a test measure the same construct.

Inter-rater reliability- this evaluates the agreement between different observers or raters when assessing the same subject or situation.

To establish the reliability of the food awareness and recognition assessment tool, the test-retest method was employed for the present study. The tool was administered to the group of two children with multiple disabilities on two different occasions, separated by a time gap of one week, the results both tests were compared across four domains: cognitive, affective, psychomotor and application. The reliability of the current research tool was 1.0 indicating the consistency the tool with a great reliability score on the Cronbach Alpha.

Phase 5: Final Tool Structure

After revision, the final tool included 20 items- scored on a simple 3-point rating scale. This format was chosen to facilitate ease of use of both parents and professionals.

Final structure of the tool

Domain	Cognitive	Affective	Psychomotor	Application
No of items	6	4	6	4
Rating scale	Achieved -3, Emerging -2, NA-1	Achieved -3, Emerging -2, NA-1	Achieved -3, Emerging -2, NA-1	Achieved -3, Emerging -2, NA-1

Maximum score	18	12	18	12
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Overall score - 60

Figure 2 -Final structure of Food Awareness and Recognition Assessment Tool

3.12 Data Collection

The data collection process was carried out in a systematic and ethical manner, following research protocols.

3.12.1. a) Permission and Consent:

Official permission was obtained from the Head of the Model School under the National Institute for Empowerment of Persons with Multiple Disabilities (NIEPMD), Chennai.

Informed consent was taken from the parents/guardians, teachers of all participating children. The purpose of the study, confidentiality, and voluntary participation were explained in detail using a consent form in understandable language.

3.12.2 b) Administration of the tool

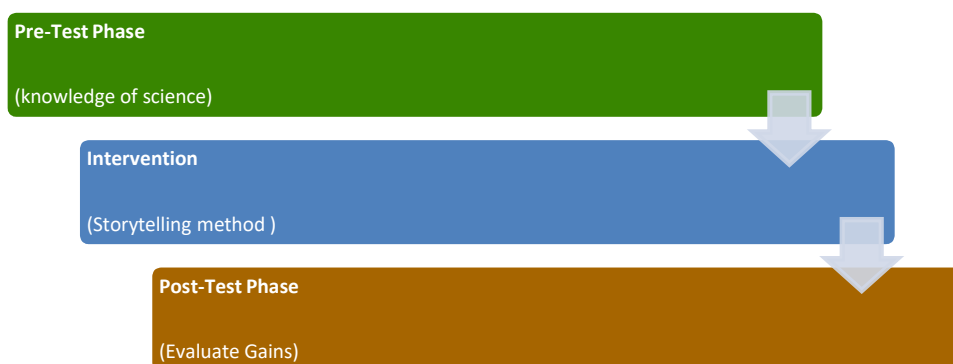
The tool developed for assessing communication domains (Cognitive, Affective, Psychomotor and Application) was administered through multiple modes to ensure a holistic and accurate evaluation:

i. **Parental input:** Parents completed the assessment tool based on their daily observations of the child's activity at home.

ii. **Teacher feedback:** Special educators the assessment tool based on classroom interactions and activities of the students.

iii **Direct Observation and Interview:** The researcher conducted informalinterviews and direct observations of each child in different settings (e.g.. classroom, playtime) to cross validate the responses.

3.13 Testing and intervention



3.13.1 Topic Selected for the Intervention

The science topic selected for the intervention was **Food**.

The topic included the following concepts:

- Healthy and unhealthy foods
- Fruits and vegetables

- Importance of balanced diet
- Energy-giving foods
- Body-building foods
- Good eating habits

The topic was selected because it is closely related to the daily lives of students and provides opportunities for concrete learning experiences through storytelling and real objects.

3.13.2 Materials Used for the Intervention

The following teaching-learning materials were used during the storytelling intervention:

Real Objects

Apple, Banana, Orange, Mango, Tomato, Potato, Carrot, Onion, Chips , Soft drink

Visual Materials

Flashcards of fruits and vegetables, Picture charts, Story illustrations, Food posters

Teaching Aids

- Story scripts

The use of real fruits and vegetables helped students observe, touch, identify, and understand food concepts through direct experience. The stories were designed using simple vocabulary, familiar characters, repetition, and visual support to meet the learning needs of students with

multiple disabilities. The stories were reviewed by the experts including the school teachers and after the feedback, the necessary changes were made in the final version.

3.14 -Phase of Intervention

3.14.1 Pre-test phase

Pre-test was conducted to record the baseline scores before the intervention.

3.14.2 Intervention Phase

The intervention was conducted over 3 months with 3 sessions per week for each child.

Steps:

Introduction of story telling method with familiar concept in daily routines.

Expansion of storytelling method use by integrating them into day-to-day life

Reinforcement through real-life scenarios, monitoring progress and providing additional support as needed.

Particulars of the intervention

Time duration-40 minutes

3 sessions (Monday, Wednesday, Friday)

Table 2 Particulars of the intervention

ACTIVITY	DOMAIN	DETAILS
Pretest and rapport building	ALL	Baseline assessment in cognitive, affective, psychomotor and application skill
Intervention phase using storytelling method	ALL	cognitive, affective, psychomotor and application skill

3.14.3 Post test phase

In this study, post-test was employed to objectively measure the science knowledge before and after the implementation of the storytelling method. Then the collected data were coded to further statistical analysis. The SPSS (Statistical Package for Social Sciences) software is the most popular Statistical Package that is used to perform data entry and analysis was used for the study. The detailed information about the statistical analysis incorporated and the findings are presented in the next chapter.

DATA ANALYSIS AND INTERPRETATION

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

4.1 Analysis and Interpretation

Data analysis and interpretation is an essential stage of research that convert raw information into useful conclusions. Analysis consists of organizing, and applying statistical techniques to summarize and probe data, revealing trends, patterns, and relationships. Interpretation then explains the meaning of those results relative to the study's questions, hypotheses, and prior literature, weighing their implications and limitations. Rigorous analysis promotes accuracy and reliability, while thoughtful interpretation adds relevance and depth, allowing researchers to draw valid conclusions and offer evidence-based recommendations. Together, these steps strengthen a study's credibility and advance knowledge across disciplines.

The analysis and interpretation of data are fundamental components of the research process, serving as the bridge between collected data and meaningful conclusions. Data analysis involves systematically applying statistical or qualitative techniques to process, summarize, and examine raw data, uncovering patterns, relationships, and trends. Interpretation, on the other hand, focuses on explaining these findings in the context of the research questions, hypotheses, and existing literature. This stage requires critical thinking to assess the implications, limitations, and significance of the results. Proper analysis ensures accuracy and reliability, while effective

interpretation provides depth and relevance, enabling researchers to draw valid conclusions and make evidence-based recommendations, Together, these processes enhance the credibility of research and contribute to knowledge advancement in any field.

4.2 Descriptive analysis

This type summarizes and describes the main features of a data set. It uses measures like mean, median, mode, standard deviation and frequency distribution to present data in a meaningful way.

Subject	Pre test		post test	
	Raw score (60)	Percentage	Raw score (60)	Percentage
Subject 1	23	38%	30	50%
Subject 2	26	43%	33	55%
Subject 3	27	45%	32	53%
Subject 4	33	55%	41	68%
Subject 5	41	68%	47	78%
Subject 6	39	65%	46	77%

Table 3 Raw score and percentage obtained by students with multiple disabilities before and after the intervention

Table 3 presents the raw scores and percentage scores out of 60 for overall science knowledge of the six students with multiple disabilities before and after the storytelling intervention. In the pre-test, the percentage scores range from 38% (Subject 1) to 68% (Subject 5), indicating that students started with low to moderate levels of science knowledge. In the post-test, all six students show higher scores, with percentages ranging from 50% to 78%, demonstrating clear improvement after the intervention.

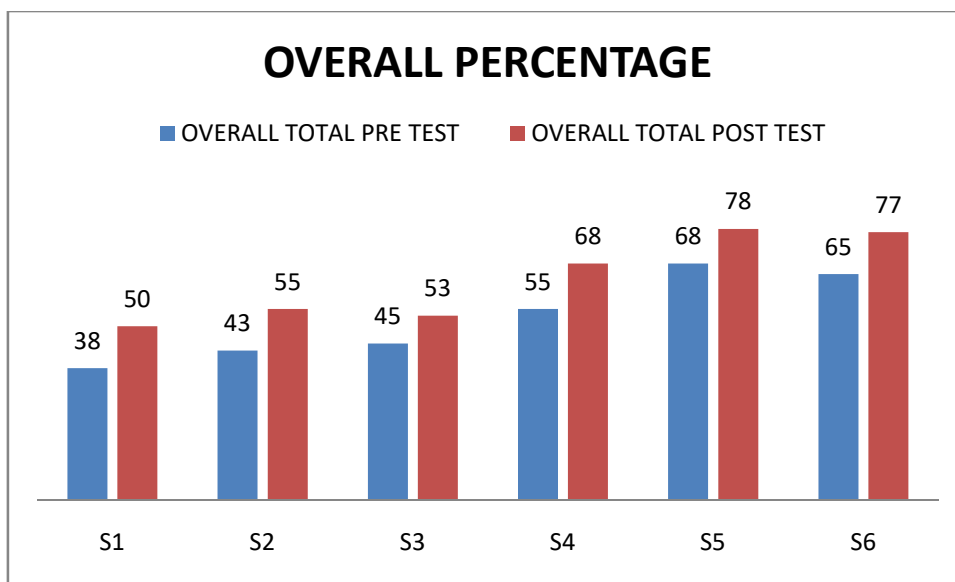


Figure 3: Graphical representation of overall percentages

Figure 3 graphically represents these pre-test and post-test percentages for each subject. The graph shows that for every student, the post-test bar is higher than the pre-test bar, visually confirming that storytelling as a teaching strategy enhanced overall science knowledge among students with multiple disabilities at the secondary level.

4.2.2 Cognitive domain scores percentage possessed by students with Multiple Disabilities before and after the intervention

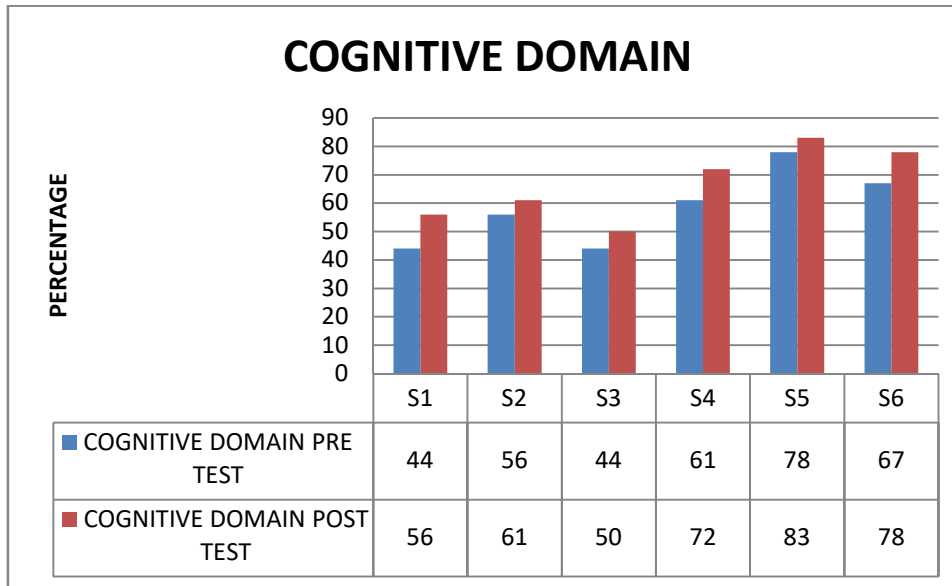


Table 4 shows the raw scores and percentages for the cognitive domain of science knowledge, recall, recognize , etc.) for each of the six students before and after the intervention. In the pre-test, students' cognitive scores range from 8 to 14, corresponding to percentage scores between 44% and 78%, indicating that some students had only basic understanding while a few showed relatively better cognitive performance. After the storytelling intervention, the post-test scores increase for all students (from 9 to 15), and the percentages rise between 50% and 83%, showing gains in cognitive processing of science concepts.

The pattern across the six students reveals consistent improvement rather than random variation. This suggests that storytelling supported better recall, understanding, and organization of basic science concepts, which are central components of the cognitive domain.

S.No	Subject	Pre test		Post Test	
		Raw Score	%	Raw Score	%
1.	S1	8	44%	10	56%
2.	S2	10	56%	11	61%
3	S3	8	44%	9	50%
4.	S4	11	61%	13	72%
5	S5	14	78%	15	83%
6.	S6	12	67%	14	78%

Table 4 :Raw score and percentage obtained by all the samples in Cognitive domain in science concepts possessed by Children with Multiple Disabilities before and after the intervention

4.2.3 Affective Domain Scores Percentage possessed by Children with Multiple Disabilities before and after the intervention

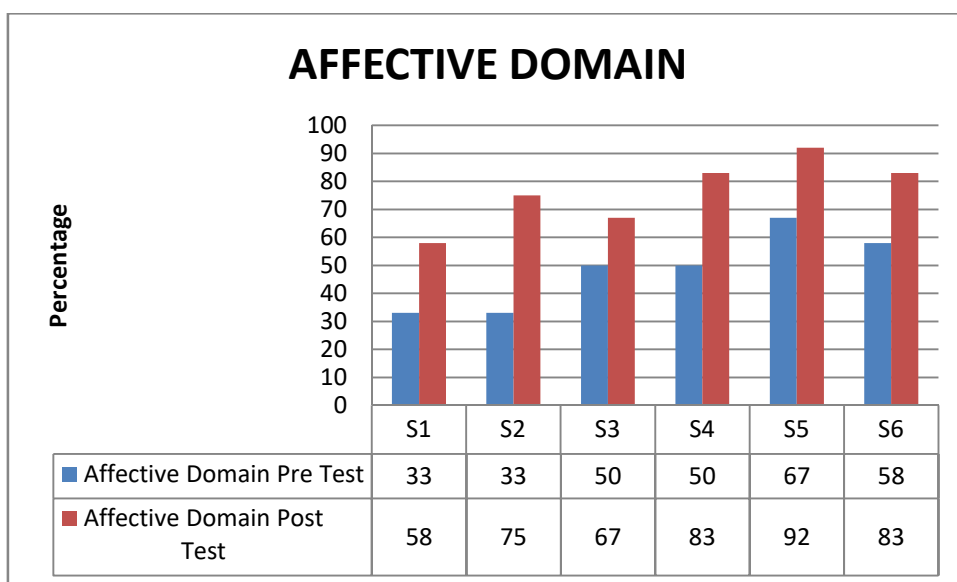


Table 4 reports the raw scores and percentages for the affective domain, which reflects interest, attitude, motivation, and emotional engagement with basic science concepts. In the pre-test, students' affective scores range from 4 to 8, corresponding to 33% to 67%, suggesting that initial attitudes and interest toward science were generally low to moderate. In the post-test, scores increase for every student, ranging from 7 to 11 with percentage scores between 58% and 92%, indicating a marked positive shift in affective engagement after the intervention.

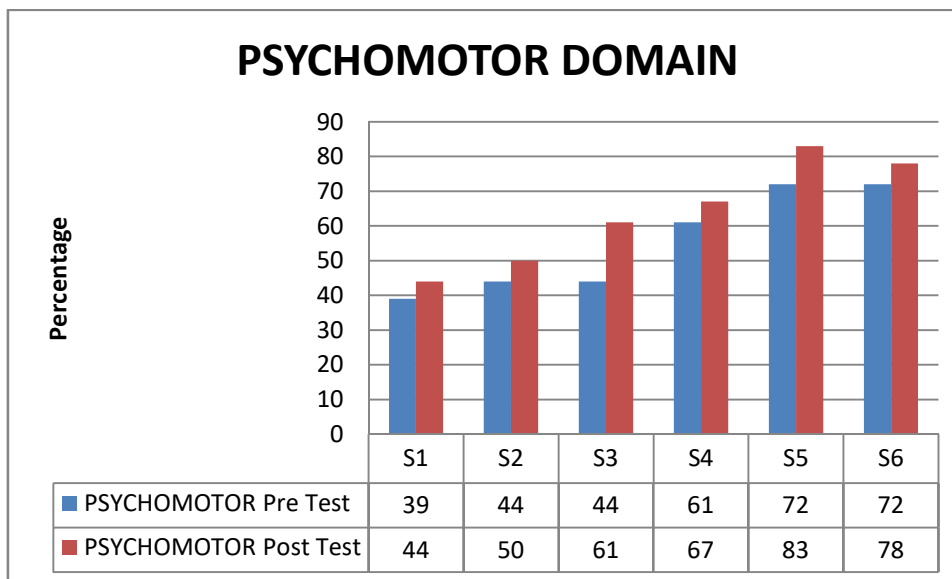
The increase is especially noticeable for students who initially had lower affective scores, such as S1 and S2, who move from 33% to 58% and 75%, respectively. This pattern suggests that storytelling made the science content more meaningful and enjoyable, thereby improving students' interest, willingness to participate, and positive feelings towards learning science.

S.No	Subject	Pre test		Post Test	
		Raw	%	Raw	%

		Score		Score	
1.	S1	4	33	7	58
2.	S2	4	33	9	75
3	S3	6	50	8	67
4.	S4	6	50	10	83
5	S5	8	67	11	92
6.	S6	7	58	10	83

Table 5 :Raw score and percentage obtained by all the samples in affective domain in science knowledge possessed by students with Multiple Disabilities before and after the intervention

4.2.4 Psychomotor Domain scores-percentage possessed by students with Multiple Disabilities before and after the intervention



S.No	Subject	Pretest	Post Test
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		Raw Score	%	Raw Score	%
1.	S1	7	39	8	44
2.	S2	8	44	9	50
3	S3	8	44	11	61
4.	S4	11	61	12	67
5	S5	13	72	15	83
6.	S6	13	72	14	78

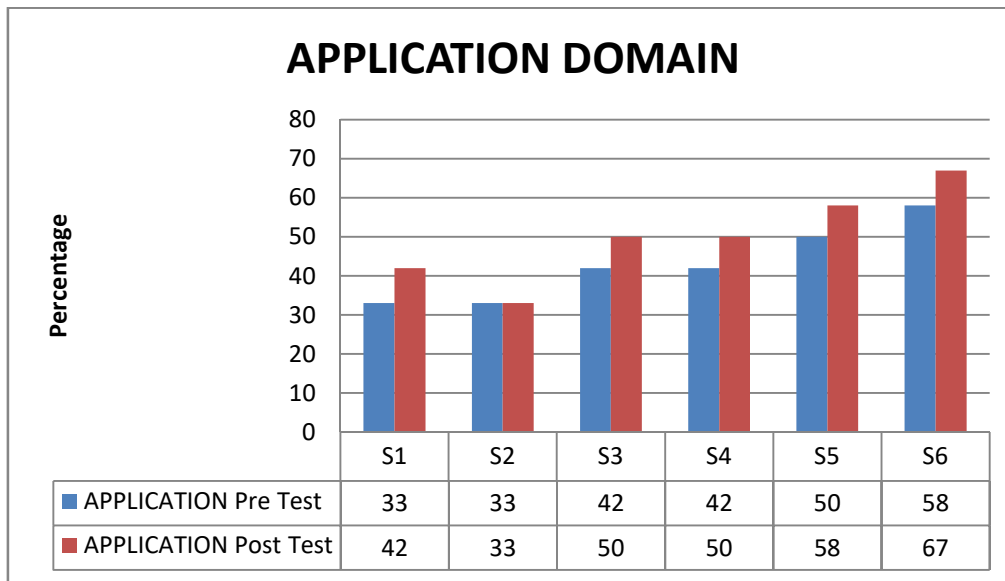
Table 6: Raw score and percentage obtained by all the samples in Psychomotor domain in science possessed by Children with Multiple Disabilities before and after the intervention

Table 6 presents the raw scores and percentages for the psychomotor domain, which relates to performance of skills, manipulation of materials, and hands-on activities in science. Pre-test psychomotor scores range from 7 to 13, with percentages between 39% and 72%, showing that students initially had limited to moderate skill performance in basic science activities. In the post-test, scores increase for all students, ranging from 8 to 15, with corresponding percentages between 44% and 83%.

The upward shift across all six students indicates that the storytelling-based activities supported better performance of science-related tasks, possibly by giving meaningful contexts and sequences to the actions required in the psychomotor domain. This improvement suggests that integrating stories with demonstrations and practice helped students with multiple disabilities to perform science skills more confidently and accurately.

4.2.5 Application Domain scores-percentage possessed by students with Multiple Disabilities before and after the intervention

Figure 7 Raw Score and Percentage for Application Domain



S.No	Subject	Pre test		Post Test	
		Raw Score	%	Raw Score	%
1.	S1	4	33	5	42
2.	S2	4	33	4	33
3	S3	5	42	6	50
4.	S4	5	42	6	50
5	S5	6	50	7	58

6.	S6	7	58	8	67
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Table 7 :Raw score and percentage obtained by all the samples in application level knowledge in science knowledge possessed by students with Multiple Disabilities before and after the intervention

The table 7 for the application domain reports pre-test and post-test scores that reflect students' ability to apply basic science concepts in practical or real-life situations. In the pre-test, raw scores range from 4 to 7, with percentages between 33% and 58%, indicating that students initially struggled to transfer theoretical knowledge into applied contexts. In the post-test, scores increase to a range of 4 to 8, with percentages from 33% to 67%; most students show clear gains, while one student (S2) maintains the same score.

Overall, the pattern for the application domain shows that storytelling helped most students to better connect concepts with everyday situations, though the magnitude of change varies across individuals. This suggests that while storytelling is effective for promoting application, some learners may require additional or differentiated support to fully generalize science concepts.

4.3 Inferential analysis

Meaning of Inferential Analysis

Inferential analysis is used to determine whether the improvement observed after the intervention is statistically significant. The pre-test and post-test scores of the same six students with multiple disabilities are compared using the paired sample t-test.

The obtained t-values in all domains are negative because the pre-test mean was taken as the first value and the post-test mean as the second value. Since post-test scores are higher than pre-test scores, the calculated difference becomes negative. The negative sign does not indicate poor performance; rather, it indicates improvement after intervention.

Comparison of mean score obtained by individual samples before and after the intervention to improves the knowledge of science among students with multiple disabilities at secondary level. The present study adopted t test to compare the statistical difference in mean score.

4.3.1 Effect of intervention in improving the Science knowledge among students with Multiple Disabilities

Hypothesis 1: The use of storytelling significantly improves the knowledge of science among students with multiple disabilities at secondary level.

Table 8 Mean, Standard Deviation, t-value of Pretest and Post Test

Domain	Category	N	Mean	SD	Df	't'-value	Significance
Knowledge of science	Pre test	6	31.50	7.369	5	15.811	Significant
	Post test		38.17	7.468			

Table 8 presents the overall comparison of pre-test and post-test scores in science knowledge among students with multiple disabilities after the storytelling intervention.

The obtained t-value was 15.811 with 5 degrees of freedom. The very high magnitude of the t-value indicates a statistically significant difference between the pre-test and post-test scores. This significant improvement demonstrates that the gains in science knowledge were not due to chance, but were the result of the storytelling intervention. Therefore, the hypothesis stating that storytelling significantly improves science knowledge among students with multiple disabilities at the secondary level is accepted.

The overall pre-test mean score was 31.50 whereas the post-test mean score increased substantially to 38.17. The increase in mean scores clearly indicates that students performed better in science after receiving instruction through storytelling. The standard deviation values for the pre-test (7.369) and post-test (7.468) indicate that the variation in students' performance remained relatively stable.

The findings further reveal that storytelling enhanced science learning across all domains, namely cognitive, affective, psychomotor, and application domains. Thus, storytelling can be considered an effective and inclusive teaching strategy for improving science knowledge among students with multiple disabilities at the secondary level.

Hypothesis 2: The use of storytelling significantly improves the knowledge of science with respect to cognitive, affective, psychomotor and application domains among students with multiple disabilities at secondary level.

Table 9 Mean, Standard Deviation, t-value of pre test and Post Test of Cognitive, Psychomotor, Affective Skills

Domain	Category	N	Mean	SD	Df	't'-value	Significance
COGNITIVE	Pre Test	6	10.50	2.345	5	6.708	Significant
	Post Test		12.00	2.366			
AFFECTIVE	Pre test	6	5.83	1.602	5	7.906	Significant
	Post test		9.17	1.472			
PSYCHOMOTOR	Pre test	6	10.00	2.683	5	4.392	Significant
			11.50	2.739			
APPLICATION	Pre test	6	5.17	1.169	5	5.00	Significant
	Post test		6.00	1.414			

Table 9 presents the mean score, standard deviation, degree of freedom, and t-value for the cognitive, affective, psychomotor and application domain of science knowledge before and after the storytelling intervention among students with multiple disabilities.

The obtained t-value was 6.708 with 5 degrees of freedom in cognitive domain. The magnitude of the t-value indicates a significant difference between the pre-test and post-test scores. Therefore, it can be interpreted that storytelling had a positive effect on the cognitive domain of science knowledge among students with multiple disabilities at the secondary level.

The pre-test mean score of the students was 10.50, whereas the post-test mean score increased to 12.00. This increase in the mean score indicates that the students demonstrated better understanding, recall, comprehension, and retention of science concepts after the intervention. The standard deviation values for the pre-test (2.345) and post-test (2.366) show that the variation among the students' scores remained relatively consistent.

Similarly in affective domain, the calculated t-value was 7.906 with 5 degrees of freedom, which shows a statistically significant difference between the pre-test and post-test scores. The large difference in mean values further confirms that storytelling was highly effective in enhancing students' affective involvement in science learning.

With respect to psychomotor domain, The obtained t-value was 4.392 with 5 degrees of freedom, showing a significant difference between the pre-test and post-test scores. This indicates that the storytelling intervention positively contributed to the development of psychomotor skills among students with multiple disabilities. The mean score in the psychomotor domain increased from 10 in the pre-test to 11.50 in the post-test. This increase indicates improvement in students' ability to perform science-related skills and activities. The standard deviation values of 2.683 and 2.739 suggest that the distribution of scores among students remained fairly similar in both tests.

The calculated t-value in application domain was 5.0 with 5 degrees of freedom, indicating a statistically significant difference between the pre-test and post-test scores. This demonstrates that storytelling had a positive impact on students' ability to apply science knowledge. The pre-test mean score in the application domain was 5.17, whereas the post-test mean score increased to 6.00. This increase indicates that students improved in their ability to apply science concepts in practical and real-life situations after the intervention. The standard deviation values of 1.169 and 1.414 show moderate variation in the scores of the students.

Hence the hypothesis "the use of storytelling significantly improves the knowledge of science with respect to cognitive, affective, psychomotor and application domains among students with multiple disabilities at secondary level is true.

The findings suggest that storytelling enhanced students' ability to understand and remember science concepts in a more meaningful and organized manner. Hence, the storytelling intervention was effective in improving the cognitive, affective, psychomotor and application aspects of science learning. This improvement indicates that storytelling positively influenced students' interest, attitude, motivation, and emotional engagement toward science learning.

4.4 Conclusion

The findings of this study show that students with multiple disabilities at the secondary level initially possessed low to moderate levels of science knowledge across cognitive, affective, psychomotor, and application domains. After the storytelling-based intervention, all six students showed improvement in their science performance, with increases in raw scores and percentage scores in each domain as well as in overall science knowledge.

The descriptive results, supported by the inferential analysis (t-test), indicate that the post-test mean scores are consistently higher than the pre-test mean scores across all domains. The obtained t-values provide statistical evidence that these gains are not due to chance but are associated with the intervention. These results support the hypothesis that storytelling significantly improves science knowledge among students with multiple disabilities at the secondary level.

Overall, the data suggest that storytelling is not only effective for enhancing cognitive understanding of science concepts but also for strengthening students' interest and attitudes (affective), skill performance (psychomotor), and ability to apply concepts in practical situations (application). Story-based science instruction therefore appears to be a promising inclusive pedagogy for learners with multiple disabilities, helping to make abstract scientific ideas more concrete, meaningful, and accessible.

SUMMARY OF FINDINGS AND CONCLUSION

CHAPTER – V

SUMMARY OF FINDINGS AND CONCLUSION

The study focused on examining the effectiveness of storytelling as a tool to enhance science knowledge among students with multiple disabilities at the secondary level. The study investigated the science knowledge regarding the concept of food among students with multiple disabilities at secondary levels and examined the effectiveness of storytelling intervention in improving science knowledge among them. The intervention was implemented through structured storytelling sessions designed to teach science concepts in an engaging and meaningful manner.

This chapter presents the findings based on the objectives and hypotheses of the study. It also discusses how the findings relate to the review of related literature. Further, recommendations for educational practice, suggestions for future research, and concluding remarks are presented.

5.1 Objectives Of The Study

1. To analyse the science knowledge among students with multiple disabilities at the secondary level.
2. To develop and implement storytelling sessions for teaching science to students with multiple disabilities at the secondary level.
3. To check the effectiveness of storytelling in enhancing science knowledge among students with multiple disabilities at the secondary level.

5.2 Hypotheses

- The use of storytelling significantly improves the knowledge of science among students with multiple disabilities at secondary level.
- The use of storytelling significantly improves the knowledge of science with respect to cognitive, affective, psychomotor and application domains among students with multiple disabilities at secondary level.

5.3 Sample

The study consisted of six students with multiple disabilities studying at the secondary level. The participants belonged to the age group of 11–14 years and were identified with A.S.D + I.D . The sample was selected through purposive sampling.

5.4 Data Collection And Analysis

The researcher personally conducted the storytelling intervention after obtaining permission from the concerned authorities. A pre-test was administered initially to assess the science knowledge of the participants. Following the implementation of storytelling sessions, a post-test was conducted to measure the effectiveness of the intervention.

The collected data were analysed using descriptive and inferential statistical techniques. Mean, Standard Deviation, Percentage Analysis, and Paired Sample t-test were employed to analyse the data and determine the effectiveness of storytelling intervention in enhancing science knowledge among students with multiple disabilities.

5.5 Major Findings

The major findings of the study entitled “Storytelling as a Tool to Enhance Science Knowledge among Students with Multiple Disabilities at Secondary Level” are presented below based on the descriptive and inferential analyses.

1. Findings Related to Overall Science Knowledge

The descriptive analysis revealed that all six students with multiple disabilities showed improvement in their overall science knowledge after the implementation of the storytelling intervention. The pre-test percentage scores ranged from 38% to 68%, indicating low to moderate levels of science knowledge before the intervention. Following the intervention, the post-test percentage scores increased and ranged from 50% to 78%, demonstrating improvement in science learning among all participants.

The inferential analysis further confirmed the effectiveness of storytelling in enhancing science knowledge. The mean score increased from 31.50 in the pre-test to 38.17 in the post-test. The obtained t-value of 15.811 was found to be statistically significant. The findings indicate that the storytelling intervention significantly improved overall science knowledge among students with multiple disabilities at the secondary level.

2. Findings Related to Cognitive Domain

The descriptive analysis of the cognitive domain revealed improvement in students’ understanding, recall, comprehension, recognition, and retention of science concepts after the storytelling intervention. The pre-test cognitive scores ranged from 44% to 78%, whereas the post-test scores ranged from 50% to 83%. Improvement was observed among all six students.

The inferential analysis showed that the mean cognitive score increased from 10.50 in the pre-test to 12.00 in the post-test. The obtained t-value of 6.708 indicated a statistically significant difference between the pre-test and post-test scores. These findings suggest that storytelling effectively enhanced the cognitive aspects of science learning and improved students' conceptual understanding of science.

3. Findings Related to Affective Domain

The descriptive analysis indicated considerable improvement in the affective domain, including interest, motivation, attitude, emotional involvement, and participation in science learning activities. The pre-test affective scores ranged from 33% to 67%, whereas the post-test scores ranged from 58% to 92%. Students who initially demonstrated lower levels of interest showed substantial improvement following the intervention.

The inferential analysis revealed that the mean affective score increased from 5.83 in the pre-test to 9.17 in the post-test. The obtained t-value of 7.906 was statistically significant. This indicates that storytelling had a strong positive effect on students' attitudes and motivation towards science learning. Among all the domains, the affective domain showed the highest improvement following the intervention.

4. Findings Related to Psychomotor Domain

The descriptive analysis of the psychomotor domain revealed improvement in students' ability to perform science-related tasks, manipulate materials, participate in activities, and demonstrate

practical science skills. The pre-test psychomotor scores ranged from 39% to 72%, while the post-test scores ranged from 44% to 83%.

The inferential analysis indicated that the mean psychomotor score increased from 10.00 in the pre-test to 11.50 in the post-test. The obtained t-value of 4.392 was found to be statistically significant. These findings indicate that storytelling, when combined with demonstrations and activity-based learning experiences, contributed positively to the development of psychomotor skills among students with multiple disabilities.

5. Findings Related to Application Domain

The descriptive analysis of the application domain revealed improvement in students' ability to apply science concepts in practical and real-life situations. The pre-test application scores ranged from 33% to 58%, whereas the post-test scores ranged from 33% to 67%. Most students demonstrated improvement in transferring science knowledge to everyday contexts, although the extent of improvement varied among individuals.

The inferential analysis showed that the mean application score increased from 5.17 in the pre-test to 6.00 in the post-test. The obtained t-value of 5.00 was statistically significant. These findings suggest that storytelling helped students connect theoretical science concepts with practical experiences and everyday situations.

6. Overall Findings of the Study

The findings of the study clearly demonstrate that storytelling was effective in enhancing science knowledge among students with multiple disabilities at the secondary level. Improvement was

observed across all domains of learning, namely cognitive, affective, psychomotor, and application domains.

The affective domain showed the greatest improvement, indicating that storytelling significantly increased students' interest, motivation, and engagement in science learning. Improvement was also observed in cognitive learning, where students demonstrated better understanding and retention of science concepts. Positive gains in psychomotor and application domains indicate that storytelling supported practical learning experiences and helped students apply scientific knowledge in real-life situations.

The statistically significant differences between pre-test and post-test mean scores across all domains provide strong evidence that the storytelling intervention was successful in enhancing science knowledge among students with multiple disabilities. Therefore, storytelling can be considered an effective, engaging, and inclusive instructional strategy for teaching science to students with multiple disabilities at the secondary level.

5.6 Discussion Of Findings

The findings are also supported by Phillips (2018), who emphasized that storytelling is a powerful instructional tool that enhances knowledge acquisition, memory recall, creativity, and learner engagement. Similar observations were made in the present study, where students showed increased interest, participation, and enthusiasm towards science learning when concepts were presented through stories.

The findings further align with the study conducted by Engel et al. (2018), which reported that storytelling significantly improved academic achievement and increased students' interest in learning among children with intellectual disabilities. Similarly, students in the present investigation demonstrated greater involvement in learning activities and improved science knowledge following the storytelling intervention. The consistency between the findings suggests that storytelling can serve as an effective instructional strategy for learners with disabilities.

The present findings are also supported by Ghaderi, Yarahmadi and Ghavami (2021), who found that storytelling improved auditory memory among children with learning disabilities. In the present study, students showed improved recall and retention of science concepts after participating in storytelling sessions. This suggests that storytelling enhances memory processes by presenting information in a structured and meaningful format.

The findings are further supported by Knight et al. (2021), who reported that instructional approaches which make abstract science concepts more accessible contribute to improved science learning among students with disabilities. Storytelling similarly helped students understand science concepts by connecting abstract ideas with familiar situations and meaningful experiences.

The consistency between the findings of the present study and previous research indicates that storytelling is an effective pedagogical approach for promoting science learning among students with multiple disabilities. The intervention not only improved understanding of science concepts but also enhanced students' interest, participation, engagement, and application of knowledge.

5.7 Recommendations For Educational Practice

1. The findings of the study indicate that storytelling is an effective strategy for teaching science concepts to students with multiple disabilities. Therefore, storytelling may be incorporated into special and inclusive classroom practices.
2. Storytelling can be used to teach other academic subjects such as mathematics, social science, environmental studies, and language learning.
3. Teachers may integrate visual aids, puppets, role-play, digital stories, and activity-based learning with storytelling to make instruction more effective.
4. Teacher education programmes may include training on storytelling techniques for inclusive education settings.

5.8 Suggestions For Further Research

1. Similar studies may be conducted with larger samples to improve the generalizability of findings.
2. Future research may investigate the effectiveness of storytelling among different disability groups and age levels.
3. Comparative studies may be undertaken to compare storytelling with other instructional strategies such as multisensory teaching, experiential learning, and technology-assisted instruction.
4. Future researchers may explore the effectiveness of digital storytelling in enhancing science learning among students with disabilities.

5. Longitudinal studies may be conducted to examine the long-term retention of science concepts acquired through storytelling.

5.9 Conclusion

The findings of the study reveal that storytelling significantly enhances science knowledge among students with multiple disabilities at the secondary level. The intervention provided meaningful learning experiences that improved students' understanding, engagement, participation, retention, and application of science concepts.

The study concludes that storytelling is an effective, inclusive, and learner-centred instructional strategy for teaching science to students with multiple disabilities. By transforming abstract concepts into meaningful narratives, storytelling helps make science learning more accessible, enjoyable, and relevant for learners with diverse educational needs. The findings therefore support the use of storytelling as a valuable pedagogical approach for enhancing science knowledge among students with multiple disabilities at the secondary level.

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